

Solution Requirements

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The following requirements define exactly what the Smart Lender solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of the system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Credit officers, financial analysts, and admin users log in to the Smart Lender portal using a registered email address and password. Passwords are hashed before storage, and sessions are managed using Flask-Login with secure, expiring session tokens.	High
2	Authorization levels	Three roles are supported: Admin (manages users, models, and system configuration), Credit Officer (submits applicant data and views individual predictions), and Financial Analyst (runs batch evaluations and exports reports). Each role is restricted to its permitted routes and actions.	High
3	External interfaces	The system connects to the trained XGBoost model and scaler (joblib .pkl files) hosted on IBM Cloud, and optionally interfaces with the bank's existing core banking system or CRM via REST API for retrieving applicant records.	Medium
4	Transactions processing	Applicant data submitted through the web form is validated, scaled using the saved StandardScaler, and passed to the prediction endpoint. Each prediction request and its result (Approved/Rejected, confidence score) is logged with a timestamp for auditability.	High
5	Reporting	The system generates individual prediction reports (decision, confidence, probability breakdown) and batch	Medium

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
		summary reports for financial analysts, including approval rate, average CIBIL score, and flagged high-risk applicants, exportable as CSV/PDF.	
6	Business rules, etc.	Applications with a CIBIL score below 500 or no credit history are automatically flagged as high-risk regardless of model output. Self-employed applicants with irregular income require an additional document verification step before final approval.	Medium
7	Compliance to laws or regulations	The system adheres to RBI data protection guidelines and the Indian Digital Personal Data Protection (DPDP) Act for handling applicant financial data, including secure storage and restricted access to personally identifiable information (PII).	High
8	Other	The system maintains a model version log so that any retrained model (Decision Tree, Random Forest, KNN, or XGBoost) can be rolled back if a newer version underperforms in production.	Low

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Predictions must be returned to the credit officer immediately after form submission, with model inference and scaling completed server-side.	Prediction response time < 3 seconds per applicant
2	Scalability	The system must support increased load during high-volume periods, such as batch evaluation of large applicant pools by financial analysts.	Supports up to 500 concurrent prediction requests
3	Security & Data Privacy	Applicant financial data (income, assets, credit history) must be encrypted in transit and at rest, with access restricted by role-based permissions.	TLS 1.2+ in transit; AES-256 encryption at rest
4	Reliability & Availability	The Flask application and underlying model service must remain available during business hours with minimal downtime for maintenance.	99.5% uptime per month
5	Usability & Accessibility	The web form must be simple enough for non-technical credit officers to use without training, with clear labels, validation messages, and result explanations.	Complies with WCAG 2.1 AA standards
6	Other	The model training pipeline and Flask codebase must be modular so that new classification algorithms can be added or the model retrained on updated datasets without restructuring the application.	Maintainability: new model integration in < 1 day